Skill Gap Analysis & Improvement Plan

Student Details

Name : SURYANARAYANA B G

Register Number : 7376231EC303

Department : ELECTRONICS AND COMMUNICATION ENGINEERING

Target Role : DATA ENGINEER / DATA ANALYST

Target Company : Deloitte/Accenture/Zoho

Skill Gap Analysis with Timeline

Technical Skills

Skill Category	Current Level	Gap Identified	Action Plan	Resources/Tools	Target Date	Remarks
SQL	Intermediate	Need advanced SQL	50 problems + 1 SQL project	LeetCode SQL	15 Feb 2026	Proof via repo
Python	Intermediate	Need pipeline building	Build ETL project	Pandas + MySQL	31 Jan 2026	Upload to GitHub
ETL	Beginner	No production pipeline	Build 2 ETL workflows	Python ETL	28 Feb 2026	End-to-end
Airflow	Beginner	No workflow orchestration	Learn DAG basics	Airflow	31 Mar 2026	1 scheduled pipeline
Cloud	Beginner	Missing	S3 + IAM + upload	AWS	30 April 2026	Include screenshots
Power BI	Intermediate	Needs advanced DAX & storytelling	Build 1 advanced dashboard with KPIs + Drill down	PowerBI	28 Feb 2026	Useful for DA shortlisting

Domain Knowledge

Category	Current Level	Gap	Action Plan	Resources	Target Completion Date	Remarks
Data Modeling	Beginner	Star schema + DWH concepts missing	Create mini data warehouse schema	Notes + YouTube	31 May 2026	Maintain documentation
Data Engineering Concepts	Beginner	Batch vs streaming, CDC, Lake/Warehouse concepts	Learn basics + apply to ETL project	Docs + tutorials	30 Apr 2026	Add notes to GitHub

Projects

Current	Gap	Action Plan	Target Date
3 projects (analyst + python)	No DE pipeline proof	Build ETL pipeline + cloud upload	June 2026
Dashboard project available	Need business storytelling dashboard	Build 1 advanced dashboard with KPIs	Feb 2026

Certifications

Current	Gap	Plan	Date
Cisco Data Analytics essentials	Need cloud cert	AWS Cloud Practitioner	Mar 2026

Soft Skills

Current Level	Gap	Plan
Intermediate	interview readiness	mock interviews weekly

Improvement Timeline

Week	Planned Activity	Topics Covered	Deliverables / Outcome
MONTH 1: Advanced SQL (Joins + Subqueries)			
Week 1	SQL Refresh + Joins	SELECT, WHERE, GROUP BY, INNER/LEFT/RIGHT JOIN	10 problems + notes
Week 2	Multi-table joins practice	Self join, multiple joins, aggregation with joins	15 problems + notes
Week 3	Subqueries (core)	Subquery types, IN/EXISTS, correlated subqueries	15 problems + notes
Week 4	Mixed practice + mini evaluation	Join + subquery mixed sets	10 problems + summary notes (1 PDF / doc)
Month 1 Total Outcome: 50 SQL problems + detailed notes			
MONTH 2: Build ETL Pipeline (CSV/API → MySQL)			
Week 1	ETL Design + DB setup	Data sources (CSV/API), DB schema design, MySQL setup	MySQL tables created + schema document
Week 2	Extract + Transform module	API calls / CSV read, cleaning (nulls/duplicates), transformation using Pandas	Python ETL script (Extract+Transform)
Week 3	Load module + SQL validation	Load into MySQL, constraints, validation queries	Final load script + SQL validation queries
Week 4	Project finishing + GitHub	README, documentation, sample input/output	Complete ETL project GitHub repo + README
Month 2 Total Outcome: GitHub ETL project + README + dataset + schema proof			
MONTH 3: Apache Airflow (Schedule ETL Pipeline)			
Week 1	Airflow Basics	DAG, tasks, operators, scheduler concept	Airflow installation + hello DAG
Week 2	Convert ETL into DAG	PythonOperator, scheduling, retries	ETL DAG running successfully
Week 3	Automation + monitoring	logs, retries, email alert basics	DAG with monitoring + logs proof
Week 4	Final pipeline + documentation	stable scheduling + documentation	Airflow scheduled pipeline + README + screenshots

Week	Planned Activity	Topics Covered	Deliverables / Outcome
Month 3 Total Outcome: DAG + pipeline automation + screenshots			
MONTH 4: AWS Basics + S3 Upload (Cloud Proof)			
Week 1	AWS Foundation	IAM, regions, buckets, security basics	AWS account setup + IAM user created
Week 2	S3 operations	upload/download, folder structure, permissions	Dataset + output uploaded to S3
Week 3	Cloud pipeline integration	Upload pipeline outputs to S3 from Python	Python script for S3 upload + proof
Week 4	Portfolio final update	Resume update + LinkedIn + GitHub update	Final resume update + cloud proof screenshots
Month 4 Total Outcome: AWS S3 proof + resume update + portfolio enhancement			

Self-Declaration

I confirm that this skill gap analysis and improvement plan is original, prepared by me, and aligned with my targeted role.

Student Signature: aprutya.b.g.

Date: 19/01/2026